This is a practice/mock assessment brief

UNIVERSITY OF LEEDS

Practice Assessment Brief

Module code & title LUBS5308M Business Analytics and Decision Science

For PRACTICE ONLY.

Module leader Aritad (Alan) Choicharoon – a.choicharoon@leeds.ac.uk

Assessment type

Assignment – 1000 Words

Learning outcomes assessed

- LO1: Critically evaluate decision science and business analytics perspectives on organisational decision support and identify and explore their complementarity
- LO2: Display familiarity with major areas of application of business analytics
- LO3: Apply accurately a range of the key methods employed in business analytic approaches to problem solving

Skills outcomes assessed

- SLO1: Apply accurately in different environments problem formulation and solving skills

Word/time limit and guidance

The word limit for this assignment is 1000 words

Write the accurate word count on the cover page. Misleading word counts can be considered academic malpractice. Please see the Assessment Policy and Procedures for further details. Remain within the word limit given above. Unless stated otherwise, it includes everything except the main title, reference list, bibliography, and appendices. Do not hide words in graphs or diagrams.

Deadline or date of assessment

Friday 13th November 2026

Submission method

Submit your assignment electronically on MINERVA by 14:00 (2pm) UK time on the deadline date. Emailed copies are not accepted. You **MUST** click the '**Submit and View Submission**' button before 14:00:00 for your assignment to be classed as submitted on time, if not your assignment will

be marked as late. It is your responsibility to ensure you upload the correct file to MINERVA, and that it has uploaded successfully.

Assessment arrangements

This is an individual assignment, and it draws on the learning in the class to cover the entire process of conducting business analytics to solve a real business problem and make decisions based on the insights.

Download and use the correct coversheet for your assignment.

Return of marks and feedback

Individual written feedback on your submission will be provided via Minerva.

Use of Assistance (including GenAI) in this assessment

The use of assistance (including GenAI assistance) is permitted for the preparation of this assessment. Please refer to the Academic Integrity and Assistance Policy for the acceptable use of assistance.

This means:

- You may use assistance (including GenAI assistance) as a tool to support you in your work, for which you remain wholly responsible. The final submission must demonstrate your own understanding, analysis, evaluation, application of theory and professional judgement.
- You remain accountable for any errors, omissions, or inaccuracies in your submission, including those produced by assistance (including GenAI assistance tools). Please be aware that the accuracy, reliability and security of technologies cannot be guaranteed.
- You must ensure that you acknowledge and reference your use of assistance (including GenAI assistance). The library website provides guidance on referencing GenAI.
- You must retain all evidence of your use of assistance (including GenAI assistance) in completing your assignment and be able to produce this evidence should your submission be investigated for academic misconduct.
- To retain evidence to demonstrate your work, you need to sign in to the assistance you use (including GenAI assistance) so you can access the history of your use if requested to do so. Please access this **infographic** so that you understand what steps you need to take to ensure you comply with what evidence you need to retain. Evidence should be retained by you until you receive confirmation of your mark for assessment by the Board of Examiners.
- Use of paid services of any assistance used to prepare assessed work is not permitted, except where approved through University disability support arrangements.

This is a practice/mock assessment brief

- See detailed assessment guidance below for any further specific instructions on the use of AI.

Note: The practice assessment does not replicate the real assessment 'size' and 'complexity' in 100%, but is intended to give you an opportunity to get some formative feedback before you tackle your assessment.

Detailed assessment guidance

Business Analytics Report (1000 words) and Technical Package: Ferngill & Co.,

1. **The background**

You have been appointed as an analytics consultant to Ferngill & Co., a speciality tea merchant founded in Windermere in 1986. Ferngill & Co. sells loose-leaf teas and subscription boxes online to customers across the UK.

The marketing director has commissioned a short evidence-based report on the company's customer base. The report should be written for a senior marketing audience: concise, well-structured, and focused on a clear, justified recommendation. The marketing director is not interested in textbook method descriptions.

1.1 The Decision

Ferngill & Co. has grown quickly, but its marketing campaigns still treat all customers as if they were the same. Personalised marketing is seen as a necessary next step to improve process efficiency.

The marketing director therefore needs to know what distinguishes Ferngill & Co.'s most valuable customers, and how the marketing team should act on that insight.

2. Your task

Task 1 — Analytics and insight (approximately 600–700 words)

Frame the business problem using CRISP-DM. Using the customer file, choose an appropriate analytical method to gain an understanding of what distinguishes Ferngill & Co.'s most valuable customers. You must:

- Clearly justify why your chosen method suits this business question and dataset,
- Identify and treat any data-quality issues you find — including the missing values — explaining your treatment and its risks,
- Interpret your findings in business terms for the marketing director,
- State explicitly what your findings imply for future marketing activity.

Conclusion - Concluding recommendation (approximately 150–200 words)

This is a practice/mock assessment brief

You must also summarise your analysis and recommend one practical action for the marketing team. State the assumptions and limitations of your analysis, and set out implementation advice.

3. Data provided

One data file has been prepared for you, and the data is about Ferngill's current customer file (`ferngill_customers.csv`)

There are 250 rows in total; each row is a customer.

Variable	Meaning
customer_id	Unique customer identifier
age	Customer age in years
region	UK region
acquisition_channel	Channel through which the customer was first acquired
tenure_months	Months since the customer's first purchase
total_orders	Number of orders in the last 12 months
average_order_value	Average order value (£) |
distinct_product_types	Number of different tea types purchased in the last 12 months
discount_sensitivity	Share of orders bought on promotion (0–1; higher means more discount-driven) |
email_engagement	Email engagement score (0–100; higher means more engaged)
satisfaction_rating	Most recent satisfaction rating (1–5; higher is better)
subscription_status	Whether the customer has an active subscription
customer_value_12m	Total spend in the last 12 months (£) |

A note on external information

This brief, together with your data file, contains everything you need. No external data collection is required or permitted. The company, the people, and the data are fictional.

This is a practice/mock assessment brief

4. Submission requirement

You are asked to submit a 1,000-word business report and a technical pack.

4.1 1,000-word business report

The word limit is strict, and you must not go over the word limit. You may submit work that is under the word limit. You must report the word count on the title page.

- Suggested structure and indicative word counts:

Section	Words
Executive summary and business problem	150–200
Task 1: Analytics and insight	600–700
Conclusion, limitations, implementation advice	150–200

- Tables, figures, references, and appendices are excluded from the word count. However, you may not include key information that is necessary to understand the answer in the appendices.

- The executive summary must state your recommendation up front.

- Do not use the report to walk through your code or reproduce textbook method descriptions. Every methodological choice must be tied to this case.

4.2 Technical pack (no word limits)

You must also submit a ZIP containing:

- your original data file,
- your cleaned data,
- your code: R script, Python notebook, or Excel workbook — whichever you used,
- your model outputs and visualisations,
- a short README stating what each file does, how to reproduce your main results, and where the key numbers in your report appear.

The report is the main output; the technical pack supports it and allows the company to evaluate the quality and accuracy of your work. Your code must run without any need for the technical team to edit them.